

Design and Implementation of a 5-DOF Pick and Place Robotic Arm

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Abstract—Automation of object handling tasks is a significant technological advancement in most of the modern world applications. This project carefully examines the design and the development of a five-degree-of-freedom (5 DOF) robotic arm which is specifically designed for automated pick-and-place operations. The main movements of the robotic arm is determined using inverse kinematics which is a mathematical process commonly used to calculate the required rotation angles of each joint based on the desired position of the end effector. The object detection system has been implemented to identify the target object and coordinate-based positioning is combined to achieve the required movements during this whole process. Mechanical and electronic simulations were also integrated and the system's future implementations will develop this system further. This research is mainly focused on the integration of robotic kinematics, mechanical design and embedded control of the robotic arm.

Index Terms—Robotic arm, 5 DOF, Kinematics, Mechanical structure, Pick-and-place, Electronic system, Micro-controller, Object detection, Challenges and implementation

I. INTRODUCTION

Robotics is an important section of engineering which focuses on the design, development and control of machines capable of performing tasks autonomously. Robots are commonly being used in modern applications that can replace humans in a number of occupations by their own structures and operations. Especially industrial robotic designs, they combine mechanical structures, electronic components, sensors, actuators and other computational techniques and are used for the industrial tasks like transporting and unloading materials, assembling, measuring, etc. Therefore, the demand for this kind of robotic designs is gradually increasing and manufacturing of the types of robots are increasing with higher precision and greater productivity.

Robotic arms are one of the robotic systems that are widely used for material handling, assembly, manufacturing and pick-and-place operations in various industries. It functions more similarly to a human arm. A robotic arm consists of connected links, joints and gripper mechanism which provide movement and pick-and-place operations within the defined workspace. In robotics, Degree Of Freedom (DOF) is defined as the number of independent axes that a robot can move. In most of the manufacturing industry, robotic arms have replaced human hands with applications like assembling, repairing and pick-and-place goods according to their assigned placements. As

mentioned before, pick-and-place is one of the fundamental applications of robotic arms in industry where an object is picked from one location and transferred to another. All joints of the robotic arm should be accurately positioned for this task to succeed. The accurate joint configurations and the coordinate system is calculated by using inverse kinematics. Inverse kinematics establishes the relationship between the desired position of the end effector and the corresponding joint angles of the robotic arm.

The purpose of this project is focused on the design and implementation of the 5 DOF robotic arm with the goal of performing an automated pick-and-place operation. This system integrates mechanical structure which designed using Solid-Works, servo motors, ESP32 microcontroller and a gripper mechanism. Considering the structure of this paper, literature review, system overview and architecture, mechanical design and prototype development, Inverse kinematics model and angle calculation, electronic system and microcontroller implementation, experimental results and discussion and finally future improvements and the references.

II. LITERATURE REVIEW

Pick and place requirement is one of the common applications of robotics in modern industries. Research has demonstrated robotic arms which are capable of detecting, grasping and transferring the objects between two different locations. One of the examples is an intelligent six-axis robotic arm which used for pick-and-place operations which is used for object recognition and demonstrated the integration of object detection with robotic manipulation. [1]

The kinematics are the main concept of the robotic arm control; it ensures the movements of the end effector and according to that the configuration of the individual joints in the system. Inverse kinematics is used to determine the joints' angles and other variables to position the end effector at the target location accurately. This concepts investigates both analytical and learning based approaches and analytical methods providing solutions based on the mathematical model of the robotic manipulator. [2]

Automated pick-and-place systems consist of various units and object detection is one of the important ones. It should identify the object correctly and determine when and where the object is available for manipulation. Previous research

has investigated camera-based object detection and machine-learning approaches. Specially vision-based systems are developed for this purpose in most modern robotic systems and provide the coordinates to the robotic arm for the pick-and-place operations. [3]

Most robotic designs are integration of object detection, kinematic computation, actuator control within embedded platforms etc. A recent six DOF robotic arm study which integrated object detection and robot control on a resource constrained microcontroller is one of the examples for the demonstration of the possibility of performing perception and manipulation using an embedded architecture. [4]

III. SYSTEM OVERVIEW AND ARCHITECTURE

The project is a five-degree-of-freedom (5-DOF) robotic arm that integrates a mechanical robotic arm structure, servo motors, an ESP32 microcontroller, a gripper mechanism and an object detection system using sensors. These components work together and the main process is to identify the target object, determine its position, move the robotic arm to the required location, pick the object and place it inside a designated container. The overall system combines mechanical design, embedded control and inverse kinematics concepts to achieve the required pick-and-place operation.

A. System Overview

This robotic arm is a 5 DOF system which consists of five joints and two links and end effector. Servo motors are used to control each joint movement and the end effector (gripper) reaches the required position to pick and place the objects within the defined workspace. The gripper mechanism is used to grasp and release the object from one place to another.

The main control unit of this robotic arm is the ESP32 microcontroller. The main purpose of it is to gather the required movement information of the five joints and generate the control signals for the servo motors and control the all joints of the robotic arm and the gripper mechanism.

The object detection in this robotic design is mainly done by the IR sensor. It identifies the arrival of the object to the defined position and according to that it determines the information required for the pick-and-place operation. The information which gets by the IR sensor will be provided to the control system. After getting this information about the object, the required joint angles using inverse kinematics based on the position of the end effector.

B. System Architecture

The overall system architecture of this robotic arm includes a few main units such as ESP32 microcontroller, servo motors and control system, IR sensor and the object detection system and the arm structure. All these units are interconnected together and work as a complete system that can perform required pick-and-place operation.

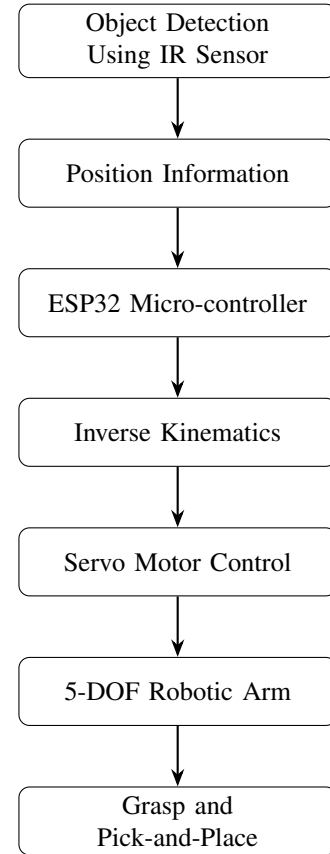


Fig. 1. Overall system flow of the proposed pick-and-place robotic arm



Fig. 2. Overall system architecture of the 5-DOF pick-and-place robotic arm

C. Working Principle

The working process of the proposed system consists of several stages. As the first step the ESP32 and the connected servo motors are initialized. The initial position of the robotic arm is predefined and the required system parameters such as angles of each joint are loaded before starting the operation.

The object detection is done by using an IR sensor at the predefined detection position. The main process is identifying the object by emitting infrared radiation and detecting the reflected signal from the object. When an object is detected by the sensor within the sensing range, it provides a corresponding signal to the ESP32 and this signal initiates the required pick-and-place operation.

Inverse kinematics is used to calculate the joint angles based on the desired position of the robotic arm. After object detection, the control algorithm determines the joint configurations and ESP32 sends the control signals to the servo motors to move correctly toward the calculated pick position.

When the end effector reaches the required object picking position which is calculated before, the gripper is activated and grasps the target object. After the object grasp, the gripper remains closed until the object is in the correct position. The gripper will securely hold the object while the robotic arm moves toward the target position and then releases the object.

After successfully picking the object the arm will move to the predefined target location which is determined using inverse kinematics calculations. The servo motors are controlled according to the joint angles and required joint configuration to place the object in the correct place by positioning the end effector correctly.

The gripper opens to release the object when the end effector reaches the target location correctly. After releasing the object in the correct position, the robotic arm returns into the predefined initial position for the next pick-and-place process.

IV. MECHANICAL DESIGN AND PROTOTYPE DEVELOPMENT

The robotic arm is designed with 5 DOF that has a working dimension which is similar to a human arm and it is capable of grasping an object along a predefined trajectory. The mechanical design of it was considered to develop a suitable structure capable of performing the pick-and-place operation. The design process initially focused on determining the required degrees of freedom, joint movements, workspace and arrangement of the robotic arm links.

A. Mechanical Design Concept

The main mechanical design concept is focused on five degrees of freedom that allow the arm to perform coordinated movements through its different joints and position the end effector at the required locations within its intended workspace. The lengths and orientations of the links were determined according to the required movements to provide sufficient reach for the intended pick-and-place task.

B. SolidWorks CAD Modeling

The three-dimensional CAD model of this robotic arm was developed using SolidWorks 2022 software. All mechanical components and parts were designed individually and then combined them to form the complete robotic arm assembly. All these links, joints and the structure were designed according to the predefined dimensions and mechanical constraints which were required for the desired application. This model provides a visual representation of the proposed mechanical structure. It was used to support the evaluation of the initial design before going to the physical prototype.

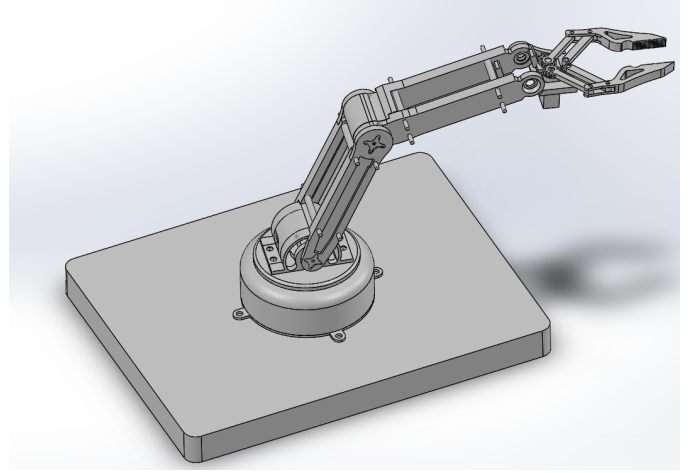


Fig. 3. 3D model of the proposed 5-DOF robotic arm

C. Final Prototype Structure Selection

A metal robotic arm structure was used for the final prototype due to the practical requirements of fabrication and implementation of the initial robotic arm structure which was designed using SolidWorks. This arm structure was selected to reduce fabrication complexity, accelerate the implementation process and increase the reliability of the system. It was adapted to accommodate the required servo motors, gripper mechanism and object detection sensor which maintains the required five-degree-of-freedom configuration.

D. Final Mechanical Assembly

The final mechanical assembly of this system consists, arm structure, servo motors, gripper mechanism and sensor mounting arrangement. Each servo motor is installed at the corresponding joints to provide controlled movement of the structure during the pick-and-place operation. Gripper mechanism is used for grasping and releasing the object in the right positions and it is mounted at the end effector. The sensor mounting system was designed to correctly detect the object while avoiding interference with the movement of the robotic arm. This final mechanical assembly was checked to ensure that the robotic arm could perform the required movements for the pick-and-place operation.

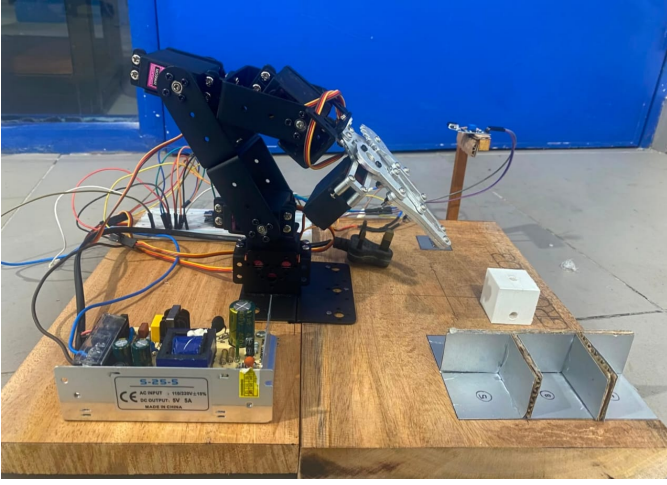


Fig. 4. Final assembled prototype of the developed robotic arm

V. INVERSE KINEMATICS MODEL AND ANGLE CALCULATION

A. Manipulator Description

The arm is a serial RRRR (revolute–revolute–revolute–revolute) manipulator with four actively driven position joints and a gripper. The wrist joint (θ_4) is held fixed throughout the pick-and-place process. Therefore, for the purpose of positioning the gripper tip, the wrist and gripper behave as a single rigid extension of the forearm. The angle θ_1 represents the base rotation angle, θ_2 represents the shoulder movement angle, and θ_3 represents the elbow movement angle.

TABLE I
JOINT CONFIGURATION OF THE ROBOTIC ARM

Joint	Name	Motion Type	Servo
1	Base	Yaw about vertical axis	DS3218
2	Shoulder	Pitch	MG996R
3	Elbow	Pitch	MG996R
4	Wrist	Pitch	MG996R
–	Gripper	Open/close	MG996R

The main link dimensions used in the kinematic model are given in Table II.

TABLE II
LINK DIMENSIONS USED IN THE KINEMATIC MODEL

Parameter	Description	Value
H	Base to shoulder effective height	100 mm
L_1	Shoulder to elbow length	100 mm
L_2	Elbow to gripper tip length	210 mm
Z	Object height	40 mm

1) Design Decision: Neutralized Wrist

Since the prototype has a small pick-and-place workspace and relatively short travel distances, wrist pitching is held at a fixed angle throughout the operation. This fixed angle

is represented by β for convenience. In the implemented prototype, $\beta = 0^\circ$, meaning that the gripper is mounted in line with the forearm.

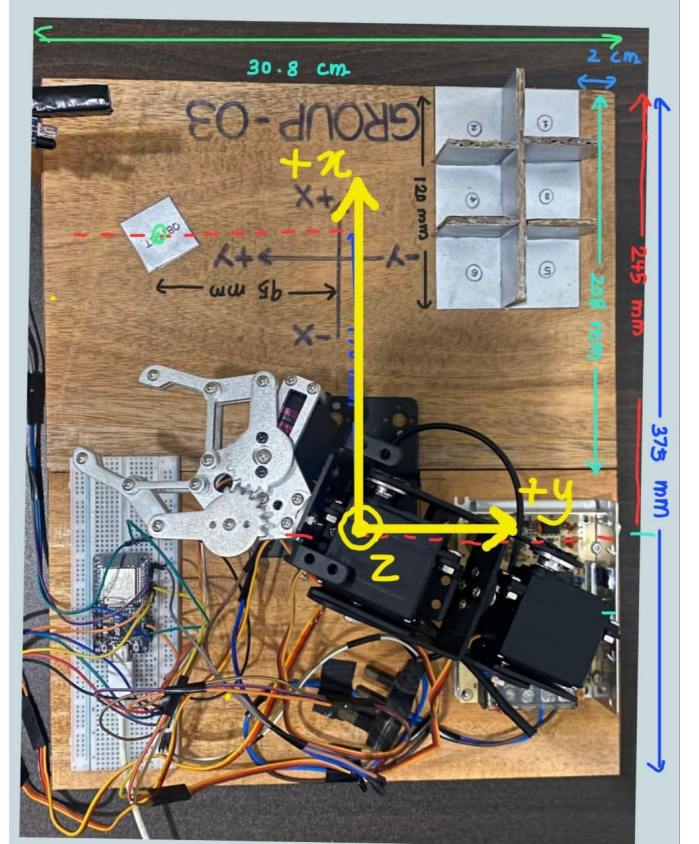


Fig. 5. Coordinate system and geometric parameters of the robotic arm for inverse kinematics analysis

2) Base Frame Convention

The base frame convention matches the calibration marks on the physical workspace. The origin is located at the intersection of the base vertical rotation axis and the table surface. The X -axis points toward the forward workspace and placement grid, the Y -axis represents the lateral direction, and the Z -axis is vertical and positive upward.

B. Denavit–Hartenberg Parameters

The standard Denavit–Hartenberg (DH) convention is used, with the vertical axis of joint 1 and parallel horizontal pitch axes for joints 2–4. The resulting DH parameters are given in Table III.

C. Explicit Joint Transformation Matrices

Using the standard DH convention, the transformation matrix for each joint is expressed as

TABLE III
DENAIVT–HARTENBERG PARAMETERS

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0°	H	θ_1
2	0	90°	0	θ_2
3	L_1	0°	0	θ_3
4	L_2	0°	0	$\theta_4 = \beta$
5 (tool)	L_g	0°	0	0

$${}^{i-1}A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -\sin \alpha_{i-1} d_i \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & \cos \alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Substituting the corresponding fixed values of α_{i-1} into the standard DH transformation gives the following explicit transformation matrices.

1) Joint 1 – Base Yaw

For $\alpha_0 = 0^\circ$ and $d_1 = H$,

$${}^0A_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & H \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2)$$

2) Joint 2 – Shoulder Pitch

For $\alpha_1 = 90^\circ$ and $a_1 = 0$,

$${}^1A_2 = \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ \sin \theta_2 & 0 & -\cos \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (3)$$

3) Joint 3 – Elbow Pitch

For $\alpha_2 = 0^\circ$ and $a_2 = L_1$,

$${}^2A_3 = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & L_1 \cos \theta_3 \\ \sin \theta_3 & \cos \theta_3 & 0 & L_1 \sin \theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (4)$$

4) Joint 4 – Wrist Pitch

The wrist joint is fixed at $\theta_4 = \beta$. Therefore, for $\alpha_3 = 0^\circ$ and $a_3 = L_2$,

$${}^3A_4 = \begin{bmatrix} \cos \beta & -\sin \beta & 0 & L_2 \cos \beta \\ \sin \beta & \cos \beta & 0 & L_2 \sin \beta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (5)$$

The transformation from the base frame to the wrist frame is therefore

$${}^0A_4 = {}^0A_1 {}^1A_2 {}^2A_3 {}^3A_4. \quad (6)$$

Since θ_4 is fixed at the constant value β , the transformation contains only three unknown joint variables, namely $\theta_1, \theta_2,$

and θ_3 . These variables can be determined from the desired gripper-tip position (X, Y, Z) without performing a separate wrist-orientation decoupling step.

D. Forward Kinematics

Multiplying the transformation chain, the gripper-tip position in the base frame can be expressed using the cumulative pitch angle

$$\phi = \theta_2 + \theta_3 + \theta_4. \quad (7)$$

The radial distance from the base axis is

$$r = L_1 \cos(\theta_2) + L_2 \cos(\theta_2 + \theta_3) + L_g \cos(\phi). \quad (8)$$

The relative vertical displacement from the shoulder axis is

$$z_{\text{rel}} = L_1 \sin(\theta_2) + L_2 \sin(\theta_2 + \theta_3) + L_g \sin(\phi). \quad (9)$$

Therefore, the Cartesian coordinates of the gripper tip are

$$X = r \cos(\theta_1), \quad (10)$$

$$Y = r \sin(\theta_1), \quad (11)$$

$$Z = H + z_{\text{rel}}. \quad (12)$$

E. Reduction Under the Neutral Wrist Assumption

Since θ_4 is held at the fixed constant β , the effective forearm length used throughout the inverse kinematics derivation is

$$L_2 = 10 + 11 = 21 \text{ cm}, \quad \beta = 0^\circ. \quad (13)$$

With $\beta = 0^\circ$, the gripper is collinear with the forearm, allowing the wrist and gripper extension to be treated as a rigid link for position calculation.

F. Inverse Kinematics (Closed Form, Elbow-Up)

Given a target gripper-tip position (X, Y, Z) in the base frame, the horizontal radial distance is calculated as

$$r = \sqrt{X^2 + Y^2}. \quad (14)$$

The base rotation angle is obtained using the two-argument arctangent function:

$$\theta_1 = \text{atan2}(X, Y). \quad (15)$$

The relative vertical position with respect to the shoulder is

$$z_{\text{rel}} = Z - H. \quad (16)$$

The distance from the shoulder joint to the target position is

$$D = \sqrt{r^2 + z_{\text{rel}}^2}. \quad (17)$$

The target position is reachable only when

$$|L_1 - L_2| \leq D \leq L_1 + L_2. \quad (18)$$

For the dimensions used in this prototype,

$$11 \text{ cm} \leq D \leq 31 \text{ cm}. \quad (19)$$

Using the law of cosines, the elbow angle is obtained from

$$\cos \theta_3 = \frac{D^2 - L_1^2 - L_2^2}{2L_1L_2}. \quad (20)$$

For the selected elbow configuration,

$$\theta_3 = -\cos^{-1}(\cos \theta_3). \quad (21)$$

The shoulder angle is then calculated as

$$\theta_2 = \text{atan2}(z_{\text{rel}}, r) - \text{atan2}(L_2 \sin \theta_3, L_1 + L_2 \cos \theta_3). \quad (22)$$

Finally, the wrist angle remains fixed at

$$\theta_4 = \beta. \quad (23)$$

G. Worked Example – Picking Position

The picking position was measured from the base rotation point as

$$X = -9.5 \text{ cm}, \quad Y = 17 \text{ cm}, \quad Z = 4 \text{ cm}. \quad (24)$$

The radial distance is

$$r = \sqrt{(-9.5)^2 + 17^2} \quad (25)$$

$$= 19.474 \text{ cm}. \quad (26)$$

The base angle is

$$\theta_1 = \text{atan2}(-9.5, 17) \quad (27)$$

$$= -29.197^\circ. \quad (28)$$

The negative value indicates that the target is located to the left of the reference direction.

The relative vertical position is

$$z_{\text{rel}} = 4 - 10 = -6 \text{ cm}. \quad (29)$$

Therefore,

$$D = \sqrt{19.474^2 + (-6)^2} \quad (30)$$

$$= 20.378 \text{ cm}. \quad (31)$$

The elbow angle is calculated using

$$\cos \theta_3 = \frac{20.378^2 - 10^2 - 21^2}{2(10)(21)} \quad (32)$$

$$= -0.2994. \quad (33)$$

Thus,

$$\theta_3 = -\cos^{-1}(-0.2994) = -107.422^\circ. \quad (34)$$

The shoulder angle is obtained from

$$\theta_2 = \text{atan2}(-6, 19.474) - \text{atan2}(21 \sin(107.422^\circ), 10 + 21 \cos(107.422^\circ)) \quad (35)$$

giving

$$\theta_2 = 62.379^\circ. \quad (36)$$

Therefore, the geometric joint angles at the pick position are approximately

$$\boxed{\theta_1 = -29.2^\circ, \quad \theta_2 = 62.4^\circ, \quad |\theta_3| = 107.4^\circ} \quad (37)$$

H. Why Calibrated Servo Angles Were Used Instead of Real-Time IK

The inverse kinematics calculation described above was performed for all seven target positions, including the picking position and six placing positions. The resulting angles were used as an initial reference for testing the robotic arm. Although the calculated values were close to the required positions, physical calibration was required to obtain accurate servo angles.

A calibration program was therefore used to manually adjust the servo positions for each target location. The final calibrated angles were then hardcoded directly into the firmware. Therefore, both the mathematical inverse kinematics model and the physical calibration process were used to obtain the final joint-angle values.

The use of calibrated angles is suitable for this prototype for the following reasons:

- The workspace is relatively small, and the angular variation between one placing slot and another is limited.
- Since all target positions are fixed, the joint angles do not need to be recalculated during every execution. The calibrated values can instead be stored directly in the firmware, reducing unnecessary ESP32 processing.
- Using calibrated servo angles reduces the possibility of commanding the robotic arm toward an unreachable position.
- The servo angles can be manually adjusted according to the actual mechanical configuration of the robotic arm rather than relying only on the ideal mathematical model.
- Since computer vision is not used for determining the target positions at this stage, repeated real-time inverse kinematics calculations are not required for the fixed target positions.

I. Comparison of Calculated Angles with Calibrated Servo Values

The same inverse kinematics procedure used for the picking position was repeated for each of the six storage slots using

their measured (X, Y) coordinates and the same vertical displacement. The calibrated picking position was used as the reference point.

The fixed offsets obtained during calibration were used to convert the calculated geometric joint angles into corresponding servo values. The offsets were

$$K = 60.803^\circ \quad (\text{base}), \quad (38)$$

$$c_1 = 60.803^\circ \quad (\text{shoulder}), \quad (39)$$

$$c_2 = 60.803^\circ \quad (\text{elbow}). \quad (40)$$

These offset values were kept unchanged for all six storage slots. The calculated joint angles and the corresponding calibrated servo values are summarized in Table IV.

1) Discussion of Results

The difference between the calculated and calibrated base angles is approximately 1° – 4° for the target positions. The close agreement indicates that the horizontal geometry, link lengths, and base rotation direction used in the model are consistent with the physical configuration. The base angle primarily depends on the horizontal position of the target and is independent of the object height.

The shoulder (θ_2) and elbow (θ_3) angles exhibit larger differences, approximately 3° – 16° , for the storage slots. These differences can be attributed mainly to practical mechanical and measurement factors rather than to the inverse kinematics formulation itself. The main contributing factors include:

- The slot coordinates were measured manually from the physical workspace, which can introduce small measurement errors.
- The actual link dimensions can differ slightly from the ideal model because of bracket thickness, servo-horn offsets, and mounting tolerances.
- The mathematical model assumes that the gripper is exactly aligned with the forearm. In practice, a small wrist or gripper offset may exist and can affect the calculated joint angles.

Accordingly, the differences between the calculated geometric angles and the calibrated servo values are mainly associated with real-world mechanical tolerances, measurement uncertainties, and the difference between the idealized kinematic model and the physical robotic arm. [5]

VI. ELECTRONIC SYSTEM AND MICROCONTROLLER IMPLEMENTATION

A. Hardware Components

The electronic system of the 5-DOF pick-and-place robotic arm consists of a microcontroller unit, servo motors, an object detection sensor, and a dedicated power supply system. The hardware components were selected to provide sufficient torque, accurate joint movement, and reliable control during the automated pick-and-place operation.

The entire system receives input on the target location, controls the robotic arm's movements through servo motors,

detects the presence of an object using an IR sensor, and then executes the required sequence of gripping and placement. The primary aim of the electronic system is to translate the calculated joint angles of the inverse kinematics algorithm into physical motion of the robotic arm.

1) ESP32 Microcontroller

The ESP32 development board was selected as the main controller due to its processing capability, multiple GPIO pins, PWM output support, and compatibility with servo motor control applications.

The ESP32 performs the following tasks:

- Generates PWM signals required for servo motor control.
- Receives input signals from the IR sensor.
- Executes the pick-and-place control algorithm.
- Stores and processes the calculated servo angle values.
- Coordinates the movement of all five servo motors.

The ESP32 acts as the communication bridge between the mathematical model and the physical robotic system. The calculated inverse kinematics results are converted into servo commands and transmitted through PWM control signals.

2) Servo Motors

The robotic arm uses five servo motors to provide five degrees of freedom. According to the system design, each joint performs a specific function, as shown in Table V.

The DS3218 servo motor was used for the base rotation due to its higher torque requirement. The MG996R servo motors were used for the remaining joints because they provide adequate torque and are commonly used for robotic arm applications.

The servo motors receive angular position commands from the ESP32 and rotate to the required positions calculated through the inverse kinematics algorithm.

3) IR Object Detection Sensor

An infrared (IR) sensor was integrated into the system to detect the presence of objects before starting the pick operation.

The sensor performs the following functions:

- Detects whether an object is available for pickup.
- Provides a trigger signal to the ESP32.
- Prevents unnecessary movement when no object is detected.

When the IR sensor detects an object, the ESP32 executes the pick-and-place sequence.

4) Power Supply System

Since multiple servo motors require high current during movement, an external power supply was used instead of directly powering the motors from the ESP32.

The power system consists of:

- 5 V external SMPS power supply.
- Breadboard-based power distribution.
- Separate servo power connections.

TABLE IV
COMPARISON BETWEEN CALCULATED JOINT ANGLES AND CALIBRATED SERVO VALUES

Position	θ_1 Calculated	θ_2 Calculated	θ_3 Calculated	Base: Calc./Code	Shoulder: Calc./Code	Elbow: Calc./Code
Pick	-29.2°	62.4°	107.4°	90.0°/90°	45.0°/45°	95.0°/95°
Slot 1	27.1°	36.7°	71.5°	33.7°/38°	19.3°/30°	59.0°/67°
Slot 2	18.4°	44.0°	82.1°	42.4°/45°	26.7°/40°	69.7°/80°
Slot 3	31.9°	52.6°	94.2°	28.9°/30°	35.2°/40°	81.7°/85°
Slot 4	22.1°	60.3°	104.7°	38.7°/40°	42.9°/50°	92.3°/100°
Slot 5	38.4°	66.5°	112.8°	22.4°/25°	49.1°/55°	100.3°/105°
Slot 6	27.3°	76.2°	124.6°	33.5°/35°	58.8°/55°	112.2°/109°

TABLE V
SERVO MOTOR CONFIGURATION

Joint	Function	Servo Motor
Joint 01	Base rotation	DS3218
Joint 02	Shoulder movement	MG996R
Joint 03	Elbow movement	MG996R
Joint 04	Wrist movement	MG996R
Joint 05	Gripper movement	MG996R

The external power supply improves system stability by reducing voltage drops and preventing controller resets caused by high servo current consumption. The project cost breakdown identifies the use of a 5 V/5 A SMPS power supply for the servo system.

B. ESP32 Control Implementation

1) Servo Library Implementation

The ESP32 program was developed using the Arduino IDE environment. The servo control system uses a servo library to generate PWM signals for controlling the angular position of each motor.

Each servo motor is assigned:

- A specific GPIO pin.
- An initial position.
- A target angle.
- A movement sequence.

The program structure consists of:

- 1) Servo initialization.
- 2) Sensor initialization.
- 3) Position definition.
- 4) Movement functions.
- 5) Pick-and-place sequence execution.

2) PWM-Based Servo Control

Servo motors are controlled using Pulse Width Modulation (PWM). The relationship between the PWM signal and servo angle is based on the following principles:

- A specific pulse width corresponds to a specific angular position.
- Changing the PWM signal changes the servo position.

The ESP32 generates PWM signals according to the required joint angles. The overall conversion process is illustrated in Fig. 6.

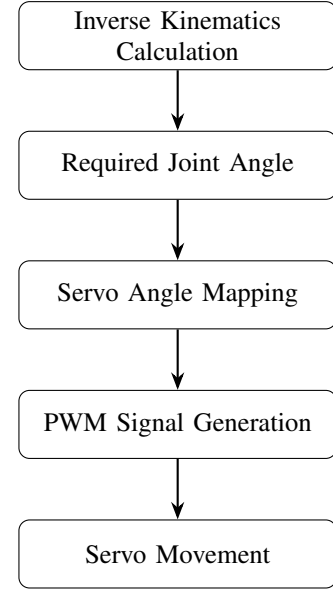


Fig. 6. Conversion of inverse kinematics results into servo movement

The calculated angles are adjusted according to the mechanical offset of each servo and converted into suitable PWM commands.

3) Software Structure

The control program was developed using the Arduino IDE environment. The software was organized into different functional sections to simplify system operation and debugging. The main program structure consists of initialization, movement control functions, and the main control loop.

a) *Initialization Section*:: During system initialization:

- Servo motors are connected to the assigned ESP32 pins.
- Initial joint positions are defined.
- The IR sensor input is configured.
- The robotic arm is moved to the home position.

b) *Movement Control Functions*:: Separate functions were developed for:

- Moving the arm to the pickup position.
- Controlling the gripper operation.
- Moving to the storage positions.
- Returning the arm to the initial position.

c) *Main Control Loop*:: The main loop continuously monitors the IR sensor status. When an object is detected, the ESP32 executes the programmed pick-and-place sequence.

C. Pick-and-Place Control Algorithm

The developed robotic arm follows a predefined automated sequence to perform object handling. The system receives a target coordinate, calculates the required joint positions, moves the arm to the required location, picks the object using the gripper, and places it into the assigned position. The operation is repeated for multiple positions within the storage box.

The complete operation consists of the following steps:

1) System Initialization

When the system is powered on, the ESP32 initializes all connected components. The servo motors move to their predefined home positions to establish a reference starting point. The initialization process ensures that the robotic arm begins every operation from a known configuration.

2) Object Detection

The IR sensor continuously monitors the workspace for object presence. When an object is detected:

- 1) The sensor sends a signal to the ESP32.
- 2) The controller processes the input.
- 3) The pick operation is initiated.

3) Movement to Pick Position

The desired object position is converted into joint angles using the inverse kinematics algorithm. The ESP32 then sends the calculated angles to the servo motors, causing the robotic arm to move through coordinated joint movements.

The movement involves:

- Base rotation adjustment.
- Shoulder positioning.
- Elbow movement.
- Wrist alignment.
- Gripper positioning.

4) Object Gripping

Once the robotic arm reaches the pickup position, the gripper servo is activated. The gripper closes around the object and maintains sufficient holding force to transport it without slipping.

5) Movement to Target Position

After successfully gripping the object, the robotic arm moves towards the assigned storage location. The movement is controlled using predefined target coordinates and corresponding servo angles.

6) Object Placement and Return

At the target location, the gripper opens and releases the object into the storage box. After placement, the robotic arm returns to the home position and waits for the next object detection signal.

VII. EXPERIMENTAL RESULTS AND DISCUSSION

A. Mechanical Testing

Mechanical testing was conducted to evaluate the stability, movement capability, and overall performance of the robotic arm structure. The main testing areas included structural stability, joint movement, and load handling capability. During development, several mechanical challenges were encountered, including arm assembly, joint alignment, load handling, and correct motor mounting.

To overcome these issues, the arm structure was reinforced, joint positions were adjusted, and a commercially available robotic arm kit was integrated as the final mechanical platform. Individual joint testing was performed to verify the following:

- Smooth base rotation.
- Accurate shoulder and elbow movement.
- Proper wrist positioning.
- Reliable gripper operation.

Servo calibration was carried out to achieve synchronized and accurate joint movement.

B. Angle Verification

The accuracy of the inverse kinematics implementation was verified by comparing the calculated joint angles with the actual servo movements. The verification process involved the following steps:

- 1) Providing the desired end-effector coordinates.
- 2) Calculating joint angles using inverse kinematics.
- 3) Sending the calculated values to the ESP32.
- 4) Observing the actual servo positions.
- 5) Adjusting calibration offsets where required.

Coordinate calibration was performed through repeated testing to improve the positioning accuracy of the robotic arm. The results confirmed that the calculated angles could be successfully converted into accurate physical movements.

C. Pick and Place Performance

The complete robotic system was tested by performing multiple pick-and-place operations. The performance evaluation included object detection, gripping performance, placement accuracy, and system repeatability.

1) Object Detection

The IR sensor successfully detected the presence of objects and triggered the automated pick-and-place operation.

2) Gripping Performance

The gripper mechanism was tested to ensure reliable object handling. Modifications were made to improve gripping performance and prevent object slipping during the pick-and-place process.

3) Placement Accuracy

Placement accuracy was improved through the following methods:

- Servo calibration.
- Coordinate adjustment.
- Repeated testing.

These adjustments helped improve the consistency of object placement at the designated positions.

4) System Repeatability

The robotic arm demonstrated repeatable movement by successfully completing multiple operating cycles. Software debugging and system integration testing were performed to ensure smooth communication between the ESP32, sensors, and servo motors. [6]

VIII. CONCLUSION

This project successfully developed a 5-DOF pick-and-place robotic arm by integrating mechanical design, inverse kinematics, electronic control, and embedded programming. A conceptual mechanical structure was designed using SolidWorks to analyse the robotic arm configuration and joint arrangement. Although the initial design was developed through CAD modeling, the final prototype was implemented using a commercially available robotic arm structure to achieve reliable mechanical operation. The inverse kinematics algorithm successfully calculated the required joint angles according to the desired end-effector positions. These calculated angles were implemented using an ESP32 microcontroller to control five servo motors and achieve coordinated robotic movement. The integration of an IR sensor enabled automated object detection, while the gripper mechanism allowed successful object handling and placement. The experimental results demonstrated that the developed system was capable of performing automated pick-and-place operations with accurate positioning and repeatable movement.

IX. FUTURE WORK

Although the developed robotic arm successfully performs basic pick-and-place operations, several improvements can be implemented to enhance system intelligence, accuracy and industrial applicability.

A. Camera-Based Object Recognition

A camera-based vision system can replace the current IR-based detection method to automatically identify the position, shape, and orientation of objects. This would allow the robotic arm to operate with different object locations without requiring predefined detection positions.

B. AI-Based Vision System

Artificial intelligence techniques can be incorporated into the vision system to enhance the autonomous capabilities of the robotic arm. The proposed AI-based system can provide the following functionalities:

- Object Classification: Identifying and categorizing different types of objects based on their visual characteristics.
- Real-Time Tracking: Continuously tracking the position of objects during the robotic operation.
- Autonomous Decision Making: Selecting appropriate actions based on the detected object and its position.

C. Feedback Control Using Encoders

Currently, the robotic arm operates using predefined servo positions. Rotary encoders can be incorporated into the joints to provide real-time position feedback. This would enable closed-loop control and improve the positioning accuracy of the robotic arm.

D. Higher Payload Design

Future improvements can include stronger mechanical structures and higher torque actuators to increase the payload capacity of the robotic arm. This would allow the system to handle heavier objects and expand its potential applications.

E. Wireless Monitoring and Control

Wireless communication technologies such as Wi-Fi or Bluetooth can be integrated into the robotic arm to enable remote monitoring and operation. This would allow users to monitor the system status and control the robotic arm without requiring a direct wired connection.

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